

Medical Image Analysis

BMED365: Topic List B01–B06

BMED365

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Overview

- 1 MRI Fundamentals
- 2 Segmentation
- 3 Quantitative Imaging
- 4 Tools for Medical Image Analysis

B01: Explain fundamental MRI principles at a high level

Magnetic Resonance Imaging (MRI):

Physical principles (simplified):

- ① **Magnetization:** Strong magnetic field aligns hydrogen atoms (protons)
- ② **RF pulse:** Radiofrequency pulse “tips” the protons
- ③ **Relaxation:** Protons return to equilibrium, emitting a signal
- ④ **Detection:** Signal is measured and reconstructed into an image

Why MRI is important in medicine:

- No ionizing radiation
- Excellent soft tissue contrast
- Many different sequences/contrasts
- Can image function (fMRI)
- Gold standard for brain, joints, spine
- Multiparametric – rich information

Relevance for AI

MRI generates large amounts of data - ideal for AI-based analysis, segmentation, and pattern recognition.

B02: Know about different MR sequences (T1, T1+Gd, T2, FLAIR)

Common MR sequences and their contrast:

Sequence	Characteristics	Clinical use
T1-weighted	CSF dark, fat bright, good anatomy	Anatomy
T1 + Gd	Contrast shows blood-brain barrier breakdown	Active tumor, metastases
T2-weighted	CSF bright, pathology often bright	Edema, inflammation, fluid
FLAIR	T2-like, but CSF dark	Lesions near ventricles, MS plaques
DWI	Diffusion of water molecules	Acute stroke (within minutes!)

For AI analysis

Multiparametric MRI (multiple sequences) provides complementary information that AI can leverage for better classification.

B03: Describe segmentation in medical image analysis

Segmentation = identify and delineate structures in images

Task:

- Assign each pixel/voxel to a **class** (e.g. tumor, normal tissue, background)
- Produces a **segmentation map** (mask)

Types of segmentation:

Semantic segmentation:

- Classify each pixel
- All tumors = one class

Instance segmentation:

- Distinguishes individual objects
- Tumor 1, Tumor 2, ...

AI methods for segmentation:

- **U-Net**: Classic encoder-decoder architecture (medical standard)
- **nnU-Net**: Self-configuring, state-of-the-art
- **Segment Anything (SAM)**: General-purpose segmentation

B04: Know about the BraTS challenge for brain tumor segmentation

BraTS = Brain Tumor Segmentation Challenge

What is BraTS?

- Annual international competition since 2012
- Benchmark for AI segmentation of brain tumors
- Multiparametric MRI (T1, T1+Gd, T2, FLAIR)

Segmentation task:

- **Whole tumor (WT):** All tumor tissue
- **Tumor core (TC):** Active + necrotic
- **Enhancing tumor (ET):**
Contrast-enhancing

Significance:

- Standardized dataset for comparison
- Drives progress in medical AI
- Foundation for clinical translation

Relevance for Team Project

BraTS is directly relevant to the glioblastoma project - students can reference BraTS architectures.

B05: Describe radiomic features and quantitative imaging

Radiomics = quantification of image characteristics

What are radiomic features?

- Mathematical description of texture, shape, intensity in images
- Hundreds of features from a single region of interest (ROI)
- “Deep” properties not visible to the human eye

Feature categories:

Intensity:

- Mean
- Standard deviation
- Histogram features

Texture (GLCM):

- Homogeneity
- Contrast
- Entropy

Shape:

- Volume
- Surface area
- Sphericity

Clinical application

Radiomics can predict molecular markers (IDH status), prognosis, and treatment response - based solely on standard images.

B06: Know about nnU-Net and MONAI as tools

nnU-Net:

- “No-new-Net” – self-configuring
- Analyzes data and selects optimal configuration
- Preprocessing, architecture, training
- State-of-the-art on many benchmarks
- Only requires data – the rest is automatic

Strengths:

- Minimal manual tuning
- Reproducible baseline

MONAI:

- Medical Open Network for AI
- PyTorch-based framework
- Specialized for medical imaging
- Transformations, networks, metrics
- Active developer community

Strengths:

- Flexibility for research
- Pre-built components

Other tools

3D Slicer: Visualization and analysis | **ITK/SimpleITK:** Image processing | **Nipype:** Neuro-pipelines

Summary: Medical Image Analysis

Key points:

- **B01:** MRI principles – magnetization, RF pulse, relaxation, detection
- **B02:** MR sequences – T1, T1+Gd, T2, FLAIR, DWI
- **B03:** Segmentation – classify each pixel, U-Net, nnU-Net
- **B04:** BraTS – benchmark for brain tumor segmentation
- **B05:** Radiomics – quantitative features for prediction
- **B06:** Tools – nnU-Net (automatic), MONAI (flexible)

Team project relevance

Medical image analysis is at the core of the glioblastoma team project. Apply your knowledge of MRI, segmentation, and radiomics.

Lab 2

Hands-on experience with CNNs for image analysis, including medical applications (MR dementia).